Bryan Zhu

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Education

University of Waterloo (Honours, Co-op) — Waterloo, ON
Bachelor of Computer Science — GPA: 4.00/4.00 — *Sep 2025 – Apr 2030*

Skills

Languages: Python, Java, C, C++, C#, TypeScript, JavaScript, HTML/CSS, SQL
Frameworks & Libraries: PyTorch, scikit-learn, ONNX, React, Node.js, Express.js, FastAPI, Electron.js, pandas, Reflex
Tools & Platforms: Git, GitHub, AWS, Docker, VS Code, MongoDB Atlas, Supabase, Chrome Extension API, SQLite

Experience

Preventative Maintenance Summer Student (Software/ML Developer) — May 2026 – Present
City of Calgary — *Calgary, AB*

- Architected and deployed a production ML pipeline auto-classifying **25,000+** municipal work orders across **20 facilities** and a 4-level taxonomy across 80+ classes, cutting manual review workload for staff by 50%+.
- Built a stacked **XGBoost + SVM ensemble** with confidence-tiered auto-fill, achieving **88% subcategory** and **86% parent category accuracy** on held-out buildings, with auto-filled predictions reaching **94-99% accuracy**.
- Closed a key generalization gap by engineering a **Gemini 2.5 Flash** fallback classifier and a taxonomy-masking architecture, lifting subcategory accuracy by up to **11 points** on unseen buildings and equipment auto-fill coverage from **39% to 65%**.

Software Project Lead — Sep 2025 – Present
Waterloo Reality Labs — *Waterloo, ON*

- **Reduced inference overhead 75%** by upgrading a real-time AR/VR hand gesture recognition system on Meta Quest from binary to multi-class dynamic classification by designing and training a **Conv2D PyTorch model** on temporal sequences.
- Achieved **99% test accuracy** across **1,246 temporal samples** with an end-to-end ML training pipeline using sliding-window augmentation (75% overlap, 20 frames), class balancing, per-feature normalization, and stratified splits.
- **Architected static and LSTM-based dynamic Siamese networks** for few-shot VR gesture recognition, redesigning the training pipeline with randomized triplet sampling, recording-level dataset splitting, and ROC-based threshold calibration.

Content Strategist (World Finalist 2024) — Nov 2023 – Apr 2024
Wharton Global High School Investment Competition — *University of Pennsylvania*

- Placed **Top 11/4000+ teams worldwide** (first-ever finalist from Alberta) by leading equity research across **20+ companies**, using macro/sector screening and fundamental analysis to develop a client-oriented investment strategy.
- Delivered a **7.11% average monthly return** by evaluating **10,000+ portfolio configurations** under client constraints and applying mean-variance optimization (efficient frontier, CML) to maximize Sharpe ratio and downside protection.

Projects

FoldForge – Differentiable Computational Origami Engine — *Python, NumPy/SciPy, PyTorch, JAX, Three.js*

- Built a **differentiable machine learning engine** from scratch, hand-deriving analytic gradients (Jacobians, sparse-Hessian implicit layer) verified to ∼**1e-8** against finite differences over a rigid-panel physics simulator.
- Cut surface-fit error **8-200x** versus a 1D baseline by training a 2D warped-Miura solver via gradient-based optimization, powering a photo-to-origami **ML pipeline** (segmentation, depth estimation, symmetry detection).
- Shipped a dependency-free, ∼**2MB** browser studio (Three.js, live physics fold solver) deployed on GitHub Pages, backed by **149 tests** across three Python versions.

VANTAGE – Autonomous Satellite Audit Platform — *Python, FastAPI, PyTorch, Reflex, Google Maps API*

- Achieved **94% validation accuracy** and 0.87 F1-score across 10 land-use classes by fine-tuning a ResNet-18 on **27,000+ EuroSAT satellite images** using transfer learning, stratified splits, and balanced sampling for deforestation risk detection.
- Maintained **60 FPS** during live audit simulation by architecting a layered Reflex geospatial dashboard with asynchronous FastAPI inference pipelines, state-managed map rendering, and Grad-CAM heatmap overlays for model explainability.
- Reduced satellite API bandwidth costs by **65%** through tile-edge caching, batched sector-level inference, and intelligent request deduplication, optimizing scan latency to **<1.2s per request** across three environmental fraud detection scenarios.

HoneyKey (nwHacks 2026) — *Python, FastAPI, HTTP Middleware, SQLite, Pydantic, Pytest, LLM Integration*

- Designed and implemented a **honeypot-based security system** to proactively detect and log real-world credential abuse and unauthorized access attempts across GitHub, client-side JavaScript, logs, and infrastructure files.
- Built a FastAPI backend with **request interception middleware** that captures **telemetry** (IP, headers, timestamps), correlates events via **SQL time-windowed queries**, synthesizes AI-generated SOC reports with multi-format firewall exports

InvestHER Financial Coach (Hack Western 12) — *TypeScript, React, Python, Supabase, Gemini*

- Built a **real-time Chrome Extension intervention engine** that detects checkout events, scrapes cart totals, injects a shadow-DOM UI across **50+ e-commerce sites**, and applies contextual spending rules to reduce impulse purchases **70%**.
- Engineered secure Supabase Edge Functions orchestrating **multi-modal AI agents** that combine Gemini for RAG-based context retrieval and ElevenLabs voice synthesis, maintaining **<800ms AI latency** under authenticated user sessions.